Atomic Physics

原子物理学

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容与

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Typeset with LaTeX

目录

1 Rutherford 模型

Coulomb 散射

$$b = \frac{a}{2}\cos\frac{\theta}{2} \qquad \left(a = r_{\min} = \frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0}E\right)$$

Rutherford 散射

$$\sigma_{\mathrm{c}}(\theta) = \frac{a^2}{16}\sin^{-4}\frac{\theta}{2} \qquad N(\theta) \propto \sin^{-4}\frac{\theta}{2}$$

散射粒子数

$$N(\theta_1 \sim \theta_2) = N_0 n t \pi \left[b(\theta_2)^2 - b(\theta_1)^2\right] = N_0 \frac{\rho N_{\mathrm{A}}}{M} t\pi \left[b(\theta_2)^2 - b(\theta_1)^2\right]$$

原子相关半径数量级

原子 $\sim 10^{-10}\,\mathrm{m}(\text{Å})$, 原子核 $\sim 10^{-14}\,\mathrm{m}$, 电子 $\sim 10^{-18}\,\mathrm{m}$.

2 Bohr 理论

Stefan-Boltzmann 定律

$$R(t) = \sigma T^4$$

Wien 位移定律

$$\lambda_{\mathrm{max}} T = b$$

光电效应

$$E_{\mathrm{k}} = h\nu = \phi = eU_{\mathrm{a}} = e(\kappa\nu - U_0)$$

$$\Longrightarrow \begin{cases} U_0 = \dfrac{\phi}{e} & \text{遏制电压} \\ h = e\kappa & \text{Planck 常量} \end{cases}$$

Rydberg 公式

$$\tilde{\nu}_{\mathrm{H}} = R_{\mathrm{H}}\left(\frac{1}{m^2} - \frac{1}{n^2}\right) = T(m) - T(n)$$

$$h\nu = hc\tilde{\nu} = \frac{hc}{\lambda} = E_n - E_m = hc[T(m) - T(n)]$$

氢原子与类氢离子的半径、速度、能级

$$r_n = \frac{n^2}{Z} a_0 \qquad v_n = \frac{Z}{n}\alpha c \qquad E_n = -\frac{Z^2}{n^2} \times 13.6\,\mathrm{eV}$$

选择定则

$$\Delta l = \pm 1$$

3 量子力学基本原理

定态 Schrödinger 方程

$$\left(-\frac{\hbar^2}{2m}\nabla^2+V\right)\psi=E\psi$$

一维无限深势阱

$$\psi(x)=\sqrt{\frac{2}{a}}\sin\frac{n\pi x}{a}\qquad E_n=\frac{n^2\hbar^2}{8ma^2}$$

一维方势垒贯穿概率

$$P=\mathrm{e}^{-2\sqrt{2m(V-E)}\Delta x/\hbar}$$

动量算符

$$\hat{p}_x\to-\mathrm{i}\hbar\frac{\partial}{\partial x}$$

力学量的平均值

$$\langle A\rangle=\int_{-\infty}^{\infty}\psi^*\hat{A}\psi\,\mathrm{d}x$$

Hermite 算符的证明

$$\int_{-\infty}^{\infty}\phi^*(\hat{F}\psi)\,\mathrm{d}x=\int_{-\infty}^{\infty}(\hat{F}\phi)^*\psi\,\mathrm{d}x$$

对易关系

$$[\hat{x}_i,\hat{x}_j]=[\hat{p}_i,\hat{p}_j]=0\qquad[\hat{x}_i,\hat{p}_j]=\mathrm{i}\hbar\delta_{ij}$$

$$\hat{L}_i=\varepsilon_{ijk}\hat{x}_j\hat{p}_k\qquad\left[\hat{L}_i,\hat{v}_j\right]=\mathrm{i}\hbar\varepsilon_{ijk}\hat{v}_k\qquad\left[\hat{L}_i,\hat{L}^2\right]=0$$

4 自旋与原子精细结构

总角动量

$$J=\sqrt{j(j+1)}\hbar\qquad J_z=m_j\hbar\qquad j=|l-s|,\cdots,l+s$$

Landé g 因子

$$g = \frac{3}{2} + \frac{1}{2}\frac{s(s+1) - l(l+2)}{j(j+1)}$$

磁矩

$$\mu_j = -g_j\sqrt{j(j+1)}\mu_{\mathrm{B}} \qquad \mu_{j,z} = -g_j m_j \mu_{\mathrm{B}}$$

$$\boldsymbol{\mu}_j = \boldsymbol{\mu}_{\mathrm{L}} \cos(\boldsymbol{L}, \boldsymbol{J}) + \boldsymbol{\mu}_{\mathrm{S}} \cos(\boldsymbol{S}, \boldsymbol{J})$$

附加能量

$$U = -m_j g_j \mu_{\mathrm{B}} B$$

Zeeman 效应

$$\Delta E = (m_2 g_2 - m_1 g_1)\mu_{\mathrm{B}} B$$

$$\Delta\nu = (m_2 g_2 - m_1 g_1)\mathcal{L} \qquad \mathcal{L} = \frac{\mu_{\mathrm{B}} B}{h}$$

$$\Delta\tilde{\nu} = (m_2 g_2 - m_1 g_1)\tilde{\mathcal{L}} \qquad \tilde{\mathcal{L}} = \frac{\mu_{\mathrm{B}} B}{hc}$$

正常 Zeeman 效应谱线的偏振

$$\Delta m_j = \begin{cases} +1 & \sigma^+ \\ 0 & \pi \\ -1 & \sigma^- \end{cases}$$

选择定则

$$\Delta S = 0 \qquad \Delta L = \pm 1 \qquad \Delta J = 0, \pm 1 \ (0 \to 0 \text{ forbidden})$$

S-G 实验

$$Z_2 = -m_j g_j \mu_{\mathrm{B}} \frac{\partial B_z}{\partial z} \frac{dD}{2E_{\mathrm{k}}}$$

P-B 效应

$$\Delta(h\nu) = (2m_s + m_l)\mu_{\mathrm{B}} B \qquad \begin{cases} \Delta m_s = 0 \\ \Delta m_l = 0, \pm 1 \end{cases}$$

5 多电子原子

原子基态

1. $S_{\max}$;
2. $L_{\max}$;
3. 未半满/半满 $J_{\min}$, 过半满 $J_{\max}$.

6 X 射线

连续谱最小波长

$$\lambda_{\min} = \frac{hc}{eV} = \frac{1.24}{V(\mathrm{kV})}\mathrm{nm} \qquad E_{\mathrm{k}} = \frac{hc}{\lambda_{\min}}$$

Moseley 定律

$$\nu_{\mathrm{K}_\alpha} = (Z-1)^2 \times 0.246 \times 10^{16}\,\mathrm{Hz}$$

$$\Delta E_{\mathrm{K}_\alpha} = (Z-1)^2 \times 13.6\,\mathrm{eV} \times \left(1 - \frac{1}{4}\right) \qquad \Delta E_{\text{K-ionization}} = (Z-1)^2 \times 13.6\,\mathrm{eV}$$

Bragg 衍射

$$n\lambda = 2d\sin\theta$$

Compton 散射

$$\Delta\lambda = \lambda_{\mathrm{C}}(1-\cos\theta) \qquad \lambda_{\mathrm{C}} = \frac{h}{m_{\mathrm{e}}c}$$

$$\Delta\lambda = \lambda_{\mathrm{C}} \quad (\theta = 90^\circ)$$

$$h\nu' = \frac{h\nu}{1+\gamma(1-\cos\theta)} \qquad \gamma = \frac{h\nu}{m_{\mathrm{e}}c^2}$$

$$E_{\mathrm{k}} = h\nu - h\nu' = \frac{hc}{\lambda} - \frac{hc}{\lambda'} = h\nu\frac{\gamma(1-\cos\theta)}{1+\gamma(1-\cos\theta)}$$

$$(h\nu')_{\min} = \frac{h\nu}{1+2\gamma} \qquad E_{\mathrm{k,max}} = h\nu\frac{2\gamma}{1+2\gamma} \qquad (\theta = 180^\circ)$$

X 射线的吸收

$$I(x) = I_0\mathrm{e}^{-\mu x}$$